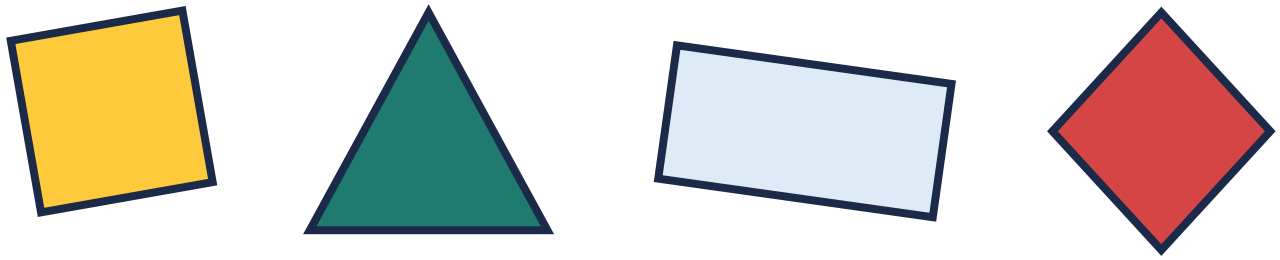
FOR PARENTS • PRIMARY 2 HOLIDAY PROGRAM

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Challenge questions if your child enjoys them.

The three levels

Core is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Digit Detectives

LESSON 1 • NUMBERS TO 1000

Core

1 What number is 5 hundreds, 3 tens and 8 ones?

2 In 914, what is the digit 1 worth?

3 What number is 100 more than 457?

4 What number is 10 less than 603?

Think Harder

5 Use the digit cards 1, 5 and 8 once each. How many different 3-digit numbers can you make? List them in order.

6 Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?

7 What number is 3 hundreds, 14 tens and 6 ones?

What your child learns

Building 3-digit numbers from hundreds, tens and ones, finding every number from digit cards by listing in order, and spotting patterns on a 1000 chart.

Tip for you

Write three digits on paper scraps. Ask your child to make the biggest and smallest numbers, then every number, in order. Before class: cut out the digit cards (kit page) and print or draw a 1000 chart with rows of 10.

Ask

"How do you know you have found them all?"

Challenge

8 Use any three of the digit cards 1, 3, 5 and 7. How many different 3-digit numbers can you make?

9 How many 3-digit numbers have digits that add up to 5?

W1 Missing Digit Mysteries

LESSON 2 • ADDITION AND SUBTRACTION WITHIN 1000

Core

1 3□5 + 214 = 579. What is the missing digit?

2 6□8 − 205 = 433. What is the missing digit?

3 What number makes this true? 400 + 250 = 300 + □

4 What must be added to 365 to make 500?

Think Harder

5 Find the missing digits: 5□7 + □48 = 905

6 Find the missing digits: 7□1 − □58 = 263

7 What number makes this true? 538 + 199 = 540 + □

What your child learns

Finding missing digits in sums and differences, checking by adding back, and magic squares with bigger numbers.

Tip for you

Cover one digit of a sum with your thumb. Let your child work out what is hiding, then check by adding. Before class: have paper scraps and a pencil ready for the number tiles 110 to 160.

Ask

"How can you check your answer?"

Challenge

8 Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as BIG as possible.

9 The numbers 101 to 109 make a magic square. What is the magic total? Which number goes in the middle?

W2 Legs, Wheels and Arrays

LESSON 3 • MULTIPLICATION AND DIVISION

Core

1 How many wheels are there on 5 bicycles?

2 There are 4 rows of chairs with 5 chairs in each row. How many chairs are there?

3 18 sweets are shared equally among 3 children. How many sweets does each child get?

4 How many legs do 7 hens have?

Think Harder

5 There are 9 hens and goats on a farm. They have 24 legs altogether. How many hens and how many goats are there?

6 Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?

What your child learns

Arrays, equal groups and sharing with leftovers, using the 2, 3, 4, 5 and 10 times tables.

Tip for you

Count legs at the dinner table or wheels in the car park. Ask how many people or cars there must be. Before class: gather about 30 counters (buttons, dried beans or Lego bricks) and paper.

Ask

"What do you do with the ones left over?"

7 An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?

Challenge

8 A shop has bicycles (2 wheels) and tricycles (3 wheels), at least one of each. There are 20 wheels altogether. Find all the ways.

9 Crayons come in boxes of 10 and packs of 5. Mei buys 6 boxes and packs altogether, with 45 crayons. How many boxes and how many packs does she buy?

W2 Pattern Machines

LESSON 4 • PATTERNS WITH MULTIPLICATION

Core

1 Count in 4s. Fill in the next three numbers: 4, 8, 12, __, __, __

2 A pattern goes red, blue, red, blue ... What colour is the 10th bead?

3 Each hand has 5 fingers. How many fingers are on 6 hands?

4 A pattern goes star, circle, circle, star, circle, circle ... What is the 9th shape?

Think Harder

5 A pattern goes A, B, B, C, A, B, B, C ... What is the 25th letter?

6 Beads go red, yellow, yellow, red, yellow, yellow ... How many yellow beads are in the first 18 beads?

What your child learns

Finding far-away places in a repeating pattern by counting in groups, and growing patterns with a table.

Tip for you

Make a pattern with spoons and forks. Ask what the 20th one will be before you build it. Before class: gather 9 red and blue cubes (or Lego bricks), about 25 lolly sticks and coloured pencils.

Ask

"How many things are in the part that repeats?"

7 Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?

Challenge

8 Ali writes 1, 2, 3, 4, 1, 2, 3, 4 ... What is the total of the first 20 numbers?

9 Houses are made of sticks in a row: 1 house uses 5 sticks, 2 houses use 9, 3 houses use 13. Ben has 40 sticks. What is the most houses he can make? How many sticks are left over?

W3 Mystery Number Machines

LESSON 5 • WORKING BACKWARDS

Core

1 I think of a number. I multiply it by 5. I get 45. What is my number?

2 I think of a number. I divide it by 2. I get 8. What is my number?

3 I think of a number. I add 35. I get 90. What is my number?

4 I think of a number. I multiply it by 3, then add 2. I get 20. What is my number?

Think Harder

5 A machine does × 4, then + 13. The OUT is 45. What went IN?

6 Ben had some money. He spent $15 on a book. Then his dad gave him $10. Now he has $42. How much money did Ben have at first?

What your child learns

Two-step number machines and working backwards from the answer, with bar models.

Tip for you

Play "I double my number and add 3, and get 15". Swap roles and let your child make up the machine. Before class: have paper strips and counters ready, and write x cards (x 2, x 3, x 4, x 5, x 10) and + or - cards (1 to 20) on small pieces of paper.

Ask

"What is the opposite of each step?"

7 Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?

Challenge

8 I think of a number. I double it, add 6, then divide by 2. I get 10. What is my number?

9 A machine does × 2, then − 4. Ravi puts a number in, then puts the OUT straight back in again. The final OUT is 12. What number did he put in first?

W3 LESSON 6 • MONEY Market Day

Core

1 Write 275¢ in dollars and cents.

2 Pay exactly 65¢ with the fewest coins.

3 Pay exactly $17 with the fewest notes.

4 A pencil costs 45¢. Mei pays with a $1 coin. How much change does she get?

Think Harder

5 How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?

6 Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.

7 A bun costs 70¢. Mei pays with a $1 coin. Which coins does she get as change, using the fewest coins?

What your child learns

Paying with the fewest notes and coins, finding every way to pay, giving change and checking a budget.

Tip for you

Let your child pay at a hawker stall or shop and count the change before you check it. Before class: cut out the play coins and notes (kit pages) and keep the spare coins in the envelope.

Ask

"Is there a way with fewer coins?"

Challenge

8 Ali has 10 coins. They are all 10¢ or 20¢ coins, worth $1.50 altogether. How many of each coin does he have?

9 Mei has $1. At the snack stall: chips 45¢, sweet 15¢, jelly 25¢, biscuit 30¢, lollipop 20¢. She wants as many DIFFERENT snacks as she can. Which should she buy?

W4 LESSON 7 • FRACTIONS
Fair Shares

Core

1 A cake is cut into 5 equal pieces. Ben eats 2 pieces. What fraction of the cake does he eat?

2 How many quarters make 1 whole?

3 Which is bigger, 1/3 or 1/6?

4 3/10 + 4/10 = ?

Think Harder

5 Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.

6 A ribbon is cut into 9 equal pieces. Mei uses 4 pieces and Siti uses 3 pieces. What fraction of the ribbon is left?

7 2/7 + ? = 1 whole

What your child learns

Halves, thirds and quarters as equal parts, comparing fractions, and adding parts of the same whole.

Tip for you

Share a roti prata or a pizza fairly. Ask if the pieces are really equal, and which is bigger: a third or a quarter. Before class: cut out the fraction strips (kit page) and have 4 paper strips of the same length.

Ask

"Are the parts the same size?"

Challenge

8 A chocolate bar has 10 equal pieces. Ali eats 1/10 of it. Mei eats 3 more pieces than Ali. What fraction of the bar is left?

9 I am a fraction with 1 on top. I am smaller than 1/4 but bigger than 1/7. What could I be? Find all the answers.

W4 LESSON 8 • 2D AND 3D SHAPES
Shape Builders

Core

1 How many quarter circles make a whole circle?

2 How many semicircles make a whole circle?

3 How many faces does a cube have?

4 How many squares of ALL sizes are in a grid 2 squares long and 2 squares high? (Small ones and one big one.)

Think Harder

5 How many quarter circles make 3 semicircles?

6 How many squares are in a grid 3 squares long and 2 squares high?

7 A stack has 2 layers. Each layer is 3 cubes long and 2 cubes wide. How many cubes?

What your child learns

Making and spotting shapes, semicircles and quarter circles, counting cubes in stacks, and nets.

Tip for you

Build a small tower with blocks or sugar cubes. Ask how many cubes there are, including the ones you cannot see. Before class: have scissors, 2 paper circles of the same size and about 12 small cubes ready, and cut out nets A to F (kit page).

Ask

"Did you count the hidden ones?"

Challenge

8 How many squares can you find in a grid 4 squares long and 3 squares high?

9 A square has both lines from corner to corner drawn. How many triangles can you find?

W5 Measure Masters

LESSON 9 • LENGTH, MASS, VOLUME AND TIME

Core

1 One bag of flour balances 2 bags of sugar. A bag of sugar is 300 g. How heavy is the bag of flour?

2 A bucket holds 10 ℓ. It has 4 ℓ of water in it. Ali pours in 3 ℓ more. How many more litres are needed to fill it?

3 A ribbon is 90 cm long. Mei cuts off 25 cm. How long is the piece that is left?

4 A cartoon starts at 5:40 p.m. and lasts 30 minutes. What time does it end?

Think Harder

5 3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?

6 You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.

What your child learns

Balance scales, pouring puzzles with jugs, lengths end to end, and telling time and how long things take.

Tip for you

Use two cups of different sizes and a basin. Can your child measure exactly one cup of water a different way? Before class: make the coat-hanger balance (hanger, two paper cups, string), have a glue stick, a spoon and 20 coins, two different-size cups with a jug of dry rice or sand (tape marks at 3 and 5), and cut out the paper clock.

Ask

"What do you know for sure from the scale?"

7 Mei's piano lesson ends at 4:15 p.m. It lasted 45 minutes. What time did it start?

Challenge

8 1 melon balances 2 pineapples. 1 pineapple balances 3 apples. 1 apple balances 2 lemons. How many lemons balance 1 melon?

9 A rope 40 cm long is cut into pieces 10 cm long. Each cut takes 2 minutes. How long does the cutting take?

W5 Challenge Day

LESSON 10 • MIXED PUZZLES

Core

1 Use the digit cards 5, 1 and 9 once each. Make the greatest 3-digit number and the smallest 3-digit number.

2 In a picture graph, each ★ stands for 2 children. Apples has 4 stars. How many children chose apples?

3 Ali has 3 bags with 5 marbles in each bag. Mei has 4 bags with 4 marbles in each bag. Who has more marbles? How many more?

4 Make 80¢ with exactly 3 coins. You can use 5¢, 10¢, 20¢ and 50¢ coins.

Think Harder

5 30 children chose a fruit. In the picture graph each ● stands for 5 children. Apple has ●● and banana has ●●●. The mango row is missing. How many ● should the mango row have?

6 How many 3-digit numbers have digits that add up to 3? List them all.

What your child learns

Choosing how to start a puzzle: draw it, list it or make a table, plus picture graphs and logic grids.

Tip for you

When your child is stuck, ask "Can you draw it?" before giving any other hint. Before class: have the digit cards, play coins, paper strips, a pencil and a small whiteboard ready.

Ask

"How did you start? What would you try next time?"

7 Ravi has 4 boxes of 5 pencils and 3 boxes of 10 pencils. He shares all the pencils equally among 5 friends. How many does each friend get?

Challenge

8 Two numbers add up to 100. One number is 30 more than the other. What are the two numbers?

9 I am a 3-digit number. My ones digit is 0. My hundreds digit is 2 more than my tens digit. My digits add up to 12. What number am I?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Digit Detectives

1. 538
2. 10 (1 ten)
3. 557
4. 593
5. 6 numbers: 158, 185, 518, 581, 815, 851.
6. 3 numbers: 447, 474, 744. The two 4s look the same, so swapping them does not make a new number.
7. 446 (14 tens = 1 hundred and 4 tens).
8. 24. Leave out one card (4 choices); each set of 3 cards makes 6 numbers: 6 + 6 + 6 + 6 = 24.
9. 15. Hundreds digit 1: 5 numbers; 2: 4; 3: 3; 4: 2; 5: 1. 5 + 4 + 3 + 2 + 1 = 15.

Week 1 • Lesson 2: Missing Digit Mysteries

1. 6 (365 + 214 = 579)
2. 3 (638 – 205 = 433)
3. 350
4. 135
5. 557 + 348 = 905 (ones 7 + 8 = 15; tens 5 + 4 + 1 = 10; hundreds 5 + 3 + 1 = 9).
6. 721 – 458 = 263. Check: 263 + 458 = 721.
7. 197. 538 is 2 less than 540, so □ is 2 less than 199. Both sides are 737.
8. 531: 654 – 123.
9. 315 (all nine add up to 945, shared by 3 rows). The middle is 105.

Week 2 • Lesson 3: Legs, Wheels and Arrays

1. 10 wheels (2, 4, 6, 8, 10).
2. 20 chairs (4 × 5 = 20).
3. 6 sweets (18 ÷ 3 = 6).
4. 14 legs (7 × 2 = 14).
5. 6 hens and 3 goats. Start with 9 hens (18 legs); each swap to a goat adds 2 legs; 3 swaps make 24. Check: 12 + 12 = 24.
6. 8 each (4 × 8 = 32) and 2 left over.
7. 16 stickers. 3 rows have 12, so each row has 4 (12 ÷ 3 = 4); 4 rows of 4 is 16.
8. 3 ways: 7 bicycles and 2 tricycles; 4 bicycles and 4 tricycles; 1 bicycle and 6 tricycles.
9. 3 boxes and 3 packs (30 + 15 = 45).

Week 2 • Lesson 4: Pattern Machines

1. 16, 20, 24
2. Blue (every even place is blue).
3. 30 fingers (5, 10, 15, 20, 25, 30).
4. A circle. 9 ends a group of 3, and each group ends with a circle.
5. A. 24 ends a group of 4 (4 × 6 = 24), so 25 starts again with A.
6. 12 yellow beads (18 beads is 6 groups of 3, with 2 yellow in each).
7. 25 sticks. The rule is add 4: 5, 9, 13, 17, 21, 25.
8. 50. 20 numbers is 5 groups, and each group adds up to 10.
9. 9 houses use 37 sticks, with 3 left over (10 houses would need 41).

Week 3 • Lesson 5: Mystery Number Machines

1. 9
2. 16
3. 55
4. 6 (20 – 2 = 18, 18 ÷ 3 = 6)
5. 8 (45 – 13 = 32, 32 ÷ 4 = 8)
6. $47 (42 – 10 = 32, 32 + 15 = 47)
7. 36 (30 – 12 = 18 is half, 18 + 18 = 36)
8. 7 (10 × 2 = 20, 20 – 6 = 14, 14 ÷ 2 = 7)
9. 6 (12 + 4 = 16, 16 ÷ 2 = 8; 8 + 4 = 12, 12 ÷ 2 = 6; check 6 → 8 → 12)

Week 3 • Lesson 6: Market Day

1. $2.75
2. 50¢ + 10¢ + 5¢ (3 coins)
3. $10 + $5 + $2 (3 notes)
4. 55¢
5. 10 ways (two $5: 1 way; one $5: 3 ways; no $5: 6 ways)
6. 2 sets: $1 + 10¢ + 10¢ + 5¢, and 50¢ + 50¢ + 20¢ + 5¢
7. 30¢: 20¢ + 10¢
8. Five 10¢ and five 20¢ (start with ten 10¢ = $1; each swap to a 20¢ adds 10¢; 5 swaps)
9. Sweet, lollipop, jelly and biscuit (15 + 20 + 25 + 30 = 90¢, 10¢ change). Five snacks cost $1.35, too much.

Week 4 • Lesson 7: Fair Shares

1. 2/5
2. 4
3. 1/3 (the same whole cut into 3 parts gives bigger parts than cut into 6)
4. 7/10
5. 1/12, 1/5, 1/3, 1/2
6. 2/9 (4 + 3 = 7 pieces used, 9 – 7 = 2 left)
7. 5/7
8. 5/10 (Ali 1 piece, Mei 4 pieces: 5 eaten, 5 left)
9. 1/5 and 1/6

Week 4 • Lesson 8: Shape Builders

1. 4
2. 2
3. 6
4. 5 (4 small + 1 big)
5. 6
6. 8 (6 small + 2 made of 4 small ones)
7. 12
8. 20 (12 + 6 + 2)
9. 8 (4 small + 4 made of 2 small ones)

Week 5 • Lesson 9: Measure Masters

1. 600 g (300 + 300)
2. 3 ℓ (4 + 3 = 7, and 10 – 7 = 3)
3. 65 cm
4. 6:10 p.m. (5:40 to 6:00 is 20 minutes, 10 more minutes)
5. Swap the pomelo for 3 apples: 5 apples balance 500 g, so one apple is 100 g. The pomelo is 300 g. Check: 300 + 100 + 100 = 500.
6. Fill the 7 ℓ jug (0, 7). Pour into the 2 ℓ jug (2, 5). Empty the 2 ℓ jug (0, 5). Pour again (2, 3). 3 ℓ is left in the 7 ℓ jug.
7. 3:30 p.m. (count back 15 minutes to 4:00, then 30 more minutes to 3:30)
8. 12 lemons (1 melon = 6 apples, and each apple = 2 lemons)
9. 6 minutes (4 pieces need only 3 cuts)

Week 5 • Lesson 10: Challenge Day

1. 951 and 159
2. 8 children
3. Mei, by 1 (Ali 15, Mei 16)
4. 50¢ + 20¢ + 10¢ (the only way)
5. 1 ● (apple 10, banana 15, so mango is 30 – 25 = 5 children)
6. 6: 102, 111, 120, 201, 210, 300
7. 10 (20 + 30 = 50 pencils; 50 ÷ 5 = 10)
8. 35 and 65 (take away the extra 30: 70, share: 35 and 35, give 30 back)
9. 750